VECTOR SPACES · INDEPENDENCE

What operations do to an independent set

Independence is fragile in one direction and robust in the other: removing vectors can never break it, adding them usually does. The two entries that read "may or may not" are the ones worth reading closely.

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OPERATIONS

UNCONDITIONAL 1

i Any subset S independent $\Rightarrow$ every $T \subseteq S$ independent — UNCONDITIONAL

REQUIRES any subset, including the empty one

Removing vectors can never create a dependence. A nontrivial combination among the survivors would have been a nontrivial combination in the original set, with zero coefficients on the removed vectors — so the original was never independent to begin with.

CONDITIONAL 3

ii Adding a vector outside the span $\mathbf{v} \notin \operatorname{span}(S) \Rightarrow S \cup \{\mathbf{v}\}$ independent — CONDITIONAL

HOLDS WHEN $\mathbf{v}$ lies outside $\operatorname{span}(S)$

A genuinely new direction extends the set safely. This is the step that builds a **basis** one vector at a time — keep adding from outside the current span until the span is everything.

v Reaching the dimension $|S| = n$ and S independent $\Rightarrow$ S is a basis — CONDITIONAL

HOLDS WHEN the space has dimension exactly n

Spanning comes free at the maximum independent size — no separate check needed. The converse shortcut also holds: n vectors that span are automatically independent. Either property plus the right count gives the other.

vi Removing from a dependent set dependence may or may not be repaired — DEPENDS WHICH

HOLDS WHEN the removed vector participates in every dependence relation

FAILS WHEN other relations survive without it

COMMON ERROR assuming any removal restores independence

Removing a vector that appears in the only relation repairs the set; removing one that does not leaves the relation intact. Which vectors are candidates is read off the **dependence relation** itself, not guessed.

BOTH OUTCOMES

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S = {(1,0), (2,0), (0,1)} — dependent, one relation
remove (2,0) → independent
remove (0,1) → still dependent: (2,0) = 2(1,0)
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FAILS OR UNDEFINED 2

iii Adding a vector inside the span $\mathbf{v} \in \operatorname{span}(S) \Rightarrow S \cup \{\mathbf{v}\}$ dependent — FAILS

FAILS WHEN $\mathbf{v}$ is any combination of S

The new vector is already expressible, and writing it as a combination and moving everything to one side gives a nontrivial relation immediately. Note this includes $\mathbf{v} = \mathbf{0}$, which lies in every span — which is why no independent set can contain the zero vector.

iv Exceeding the dimension $|S| > n$ in an n-dimensional space $\Rightarrow$ dependent — FAILS

FAILS WHEN more vectors than dimensions

No counting or computation required — the conclusion follows from the size alone. This is the fastest of all the **independence tests**, and the reason a homogeneous system with more unknowns than equations always has a nontrivial solution.

The asymmetry has a single source. Independence is a statement that no nontrivial combination gives $\mathbf{0}$, so a smaller set has fewer combinations to worry about and cannot acquire a relation it did not have. Adding a vector adds combinations, and only one of them needs to vanish for the property to be lost.